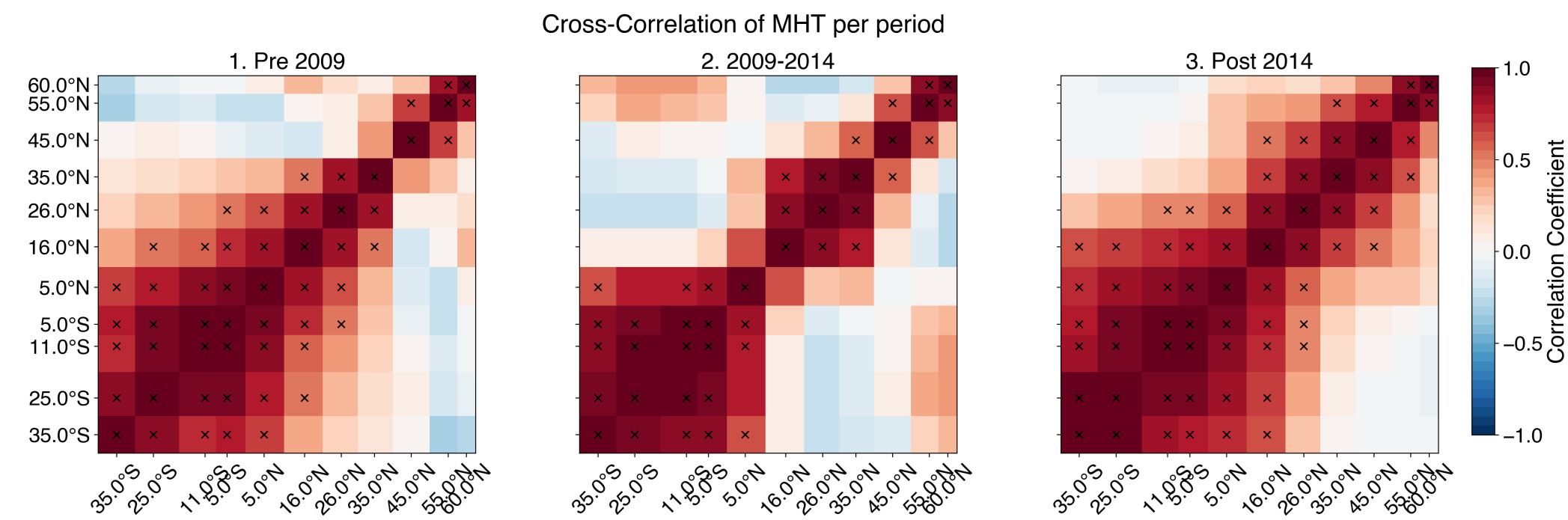


Wavelet coherence: 16°N and 5°N

Validate / investigate timescales using wavelet coherence (Cross correlation split by eye)



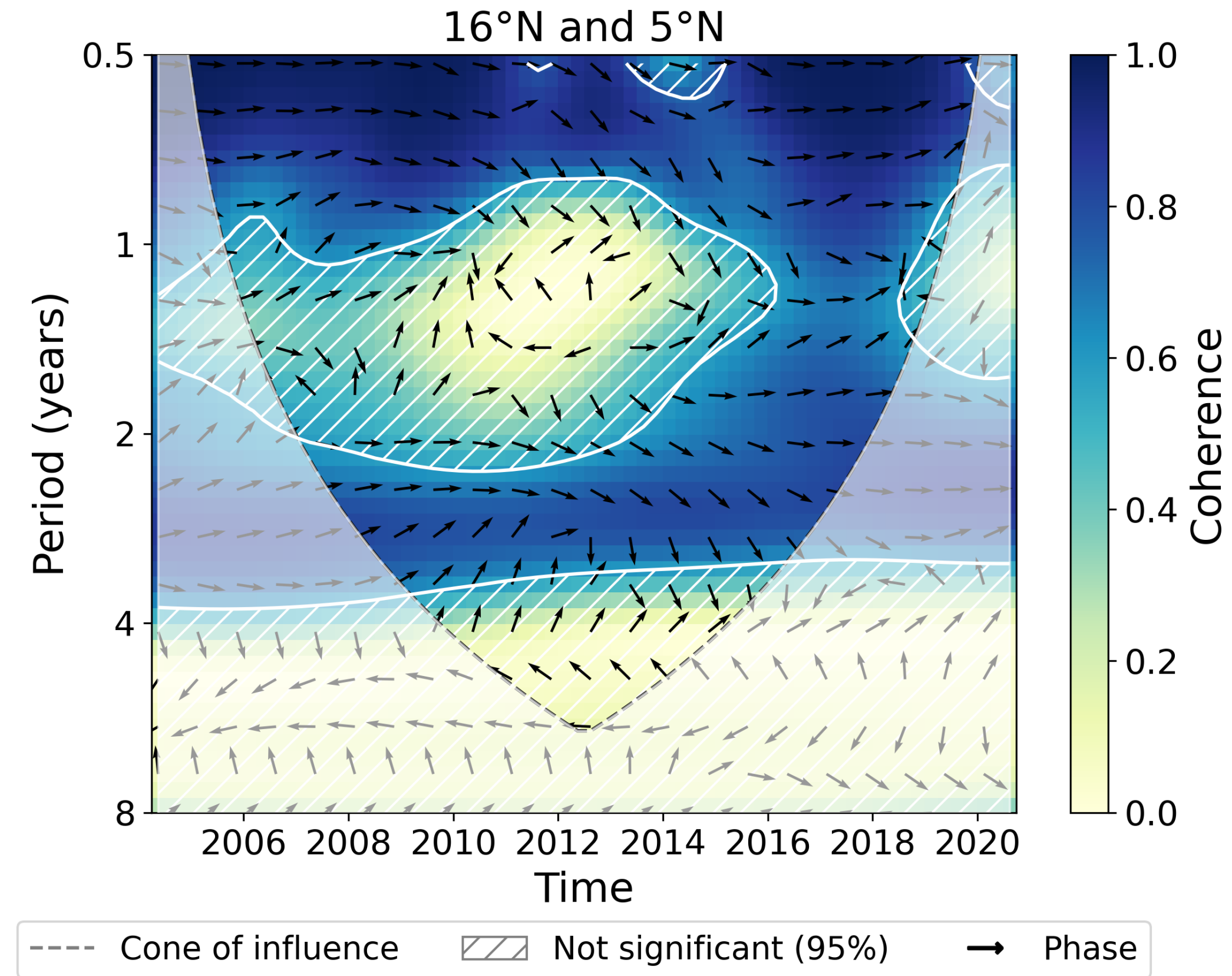
Interpretation of wavelet coherence plot:

On (bi-) annual scales:

- **Drop of coherence** below 0.4 (not significant), strongly pronounced around **2009 - 2014** (including 2014)
- Phase arrows are **scattered** around → no stable relative phase relationship

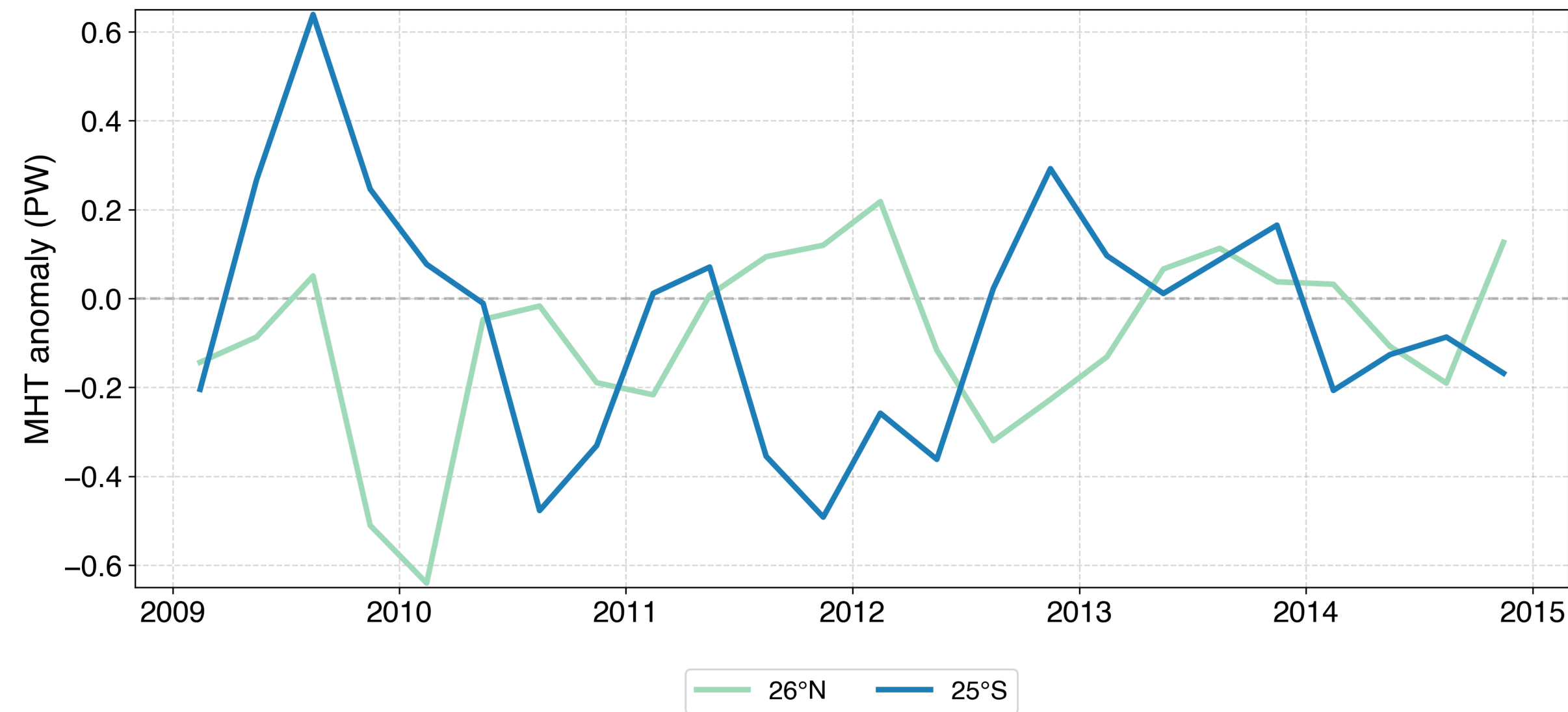
Interannual:

- Around 3 years again high coherence
- Similar drop in coherence around 4-5 year period as on annual scales (**not significant**) , out of phase (i.e. anti-correlation?)
- Around 2008 - 2010: 16°N leading by ~1 year, then around 2012 switch to 5°N leading by ~1 year

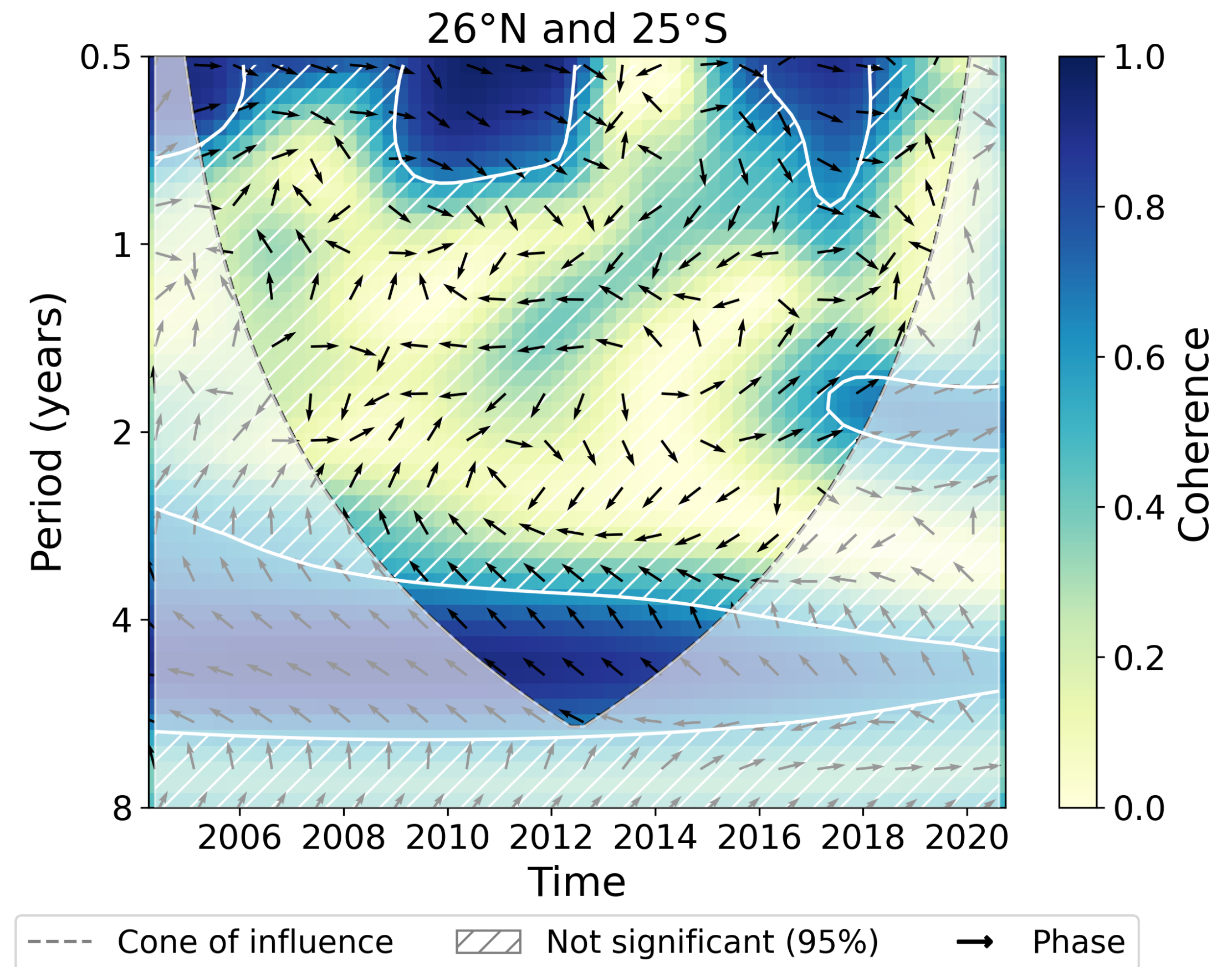


Wavelet coherence transform between 16°N and 5°N.
Period in logarithmic scale.

Wavelet coherence: 26°N and 25°S



- High coherence & out of phase around 2010-2013 (26°N leading slightly) for periods >4 years



Cross correlation between blocks shows break

- I can find the imprint of the coherence break already in the raw (and the detrended) data
 - Significance calculated for each period separately
- perform correlation analysis using the unfiltered (only detrended) data ?
- (as for wavelet coherence I should use unfiltered aswell)

Period	Corr ST & SA block	Significant (95%)
Pre 2009	0.53	Yes
2009-2014	0.01	No
Post 2014	0.47	No (rcrit: 0.48)